Umm Al-Qura University

College of Engineering and Computer Science

Computer Science Department

SE1201 – Foundations of Software Engineering

Project

Student Attendance Management System

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1. Introduction

Many educational institutions today still depend on traditional methods for recording student attendance, such as manual entry in paper sheets or spreadsheets. These methods can be time-consuming and are often prone to human errors, missing data, and difficulty in organizing or retrieving information when needed.

In this project, we designed and developed a simple and efficient system for managing student attendance. The system aims to improve the attendance process by making it more accurate, organized, and easier to use. It allows instructors to easily record attendance, enables students to check their attendance records at any time, and provides clear and organized reports for administrators to support decision-making.

This project follows the Software Development Life Cycle (SDLC) approach, starting from understanding the problem and gathering requirements, then moving through system design, implementation, testing, and finally documentation. Through this process, we were able to better understand how real software systems are planned and developed step by step.

2. Problem Statement

During the analysis of the current system, several problems were identified.

- Recording attendance manually is time-consuming, especially in large classes.
- There is a high possibility of human errors when entering or managing data.
- It is difficult to continuously track student absences and identify patterns.
- There is no quick or efficient way to generate accurate attendance reports.

In addition, traditional methods lack automation, which reduces overall efficiency and productivity. They also do not provide real-time access to attendance data, making it harder for instructors and administrators to make timely decisions. Furthermore, storing data manually increases the risk of data loss or misplacement.

These issues highlight the need for a computerized system that can improve accuracy, save time, and provide better management of student attendance.

3. Objectives

This project aims to:

- Develop an electronic system that simplifies the process of recording student attendance.
- Reduce human errors associated with manual data entry.
- Provide accurate and well-organized attendance reports.
- Facilitate the monitoring of student attendance by instructors and administrators.

In addition, the system aims to improve overall efficiency by saving time and effort, and to provide a more reliable and consistent method for managing attendance data. It also seeks to enhance decision-making by offering clear insights into student attendance patterns and behavior. Furthermore, the system is designed to be user-friendly and scalable, allowing future improvements and integration with other educational systems.

4. Planning

The planning phase is a critical step in the Software Development Life Cycle (SDLC), as it defines how the project will be carried out and completed successfully. In this project, a clear plan was developed to ensure that all tasks were organized and completed within the given time frame.

The project was divided into several phases, including proposal, requirements, design, implementation, testing, and deployment. Each phase was assigned a specific timeline using a Gantt chart to track progress and ensure that deadlines were met.

To maintain effective collaboration, regular meetings were conducted among the team members. During these meetings, different aspects of the project were discussed, tasks were distributed, and any issues or challenges were addressed. All meetings were documented in an Excel sheet, including the meeting members, discussed topics, decisions made, and the date of the next meeting.

Additionally, responsibilities were clearly assigned to each team member to ensure equal contribution and accountability. This structured planning approach helped the team stay organized, manage time efficiently, and successfully complete the project

5. Requirements

5.1 Functional Requirements

The system should:

- Allow users to log in securely using authenticated credentials.
- Enable teachers to record student attendance efficiently.
- Allow students to view their own attendance records at any time.

- Generate detailed attendance reports for classes, students, and time periods.
- Support full management of student data, including adding, updating, and deleting records.
- Provide role-based access control (e.g., admin, teacher, student).
- Ensure data security and protect user information.
- Store all records in a centralized database for easy access and consistency.
- Maintain an audit trail of attendance and system activities for accountability.
- Provide a user-friendly interface for both web and/or mobile access.

5.2 Non-Functional Requirements

The system should:

- Be fast and provide quick response time for all user actions.
- Be easy to use with a simple and intuitive user interface.
- Ensure high security to protect user data and prevent unauthorized access.
- Be scalable so it can be expanded in the future without major changes.
- Support efficient performance even when the number of users increases.
- Maintain data integrity and reliability at all times.
- Be compatible with future updates and new features.
- Provide a stable and consistent user experience across different devices.

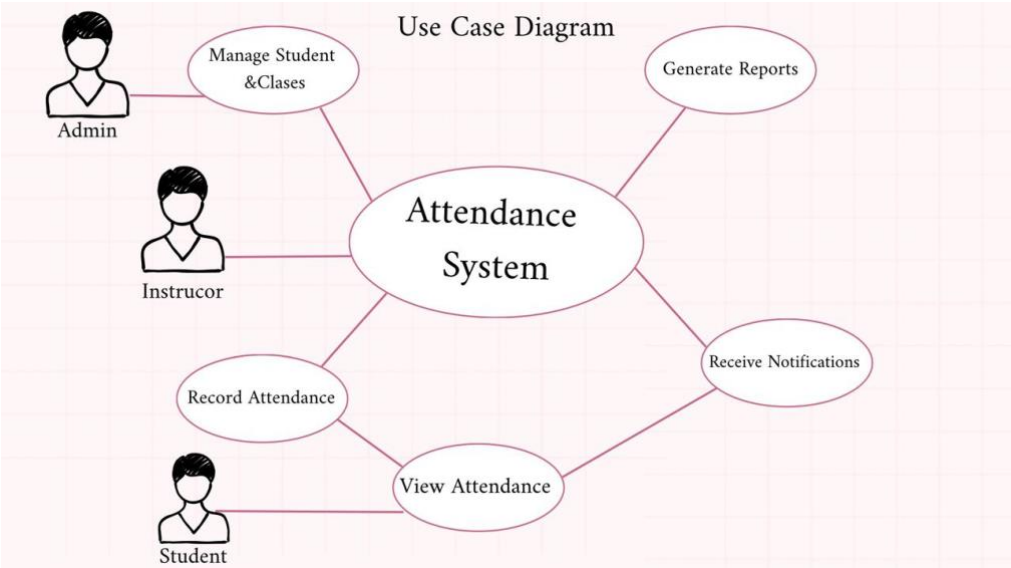
5.3 System Requirements

- A computer with suitable hardware specifications to run development tools smoothly.
- Java Development Kit (JDK) installed to support Java programming language execution.
- An Integrated Development Environment (IDE) such as Visual Studio Code (VS Code) or IntelliJ IDEA for writing and managing the code.
- A database management system (DBMS) such as MySQL or SQLite to store and manage system data.
- Required Java libraries and dependencies configured properly within the project.
- Stable operating system (Windows, macOS, or Linux) that supports Java development tools.
- Internet connection (optional) for downloading tools, updates, and dependencies.

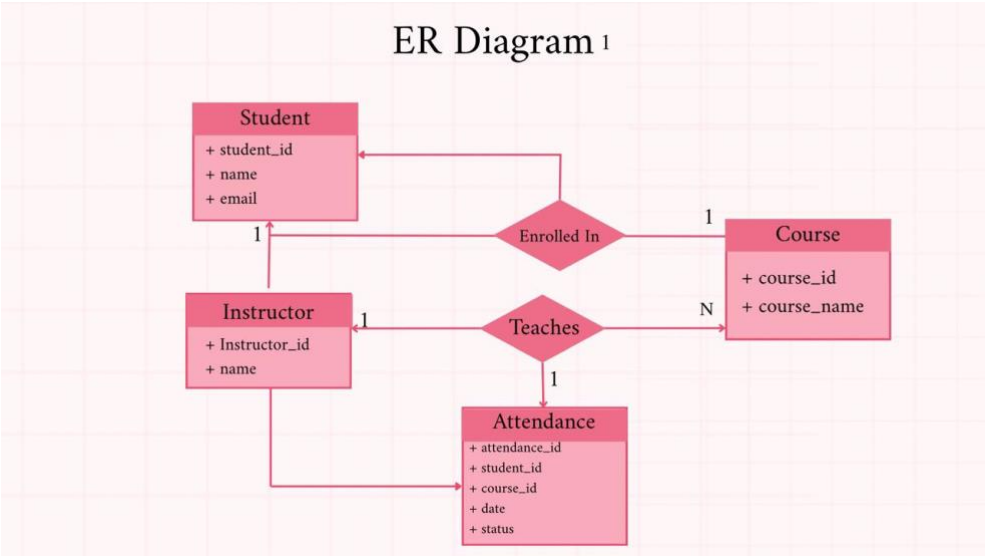
6. Design

The system was designed using several diagrams to ensure a clear understanding before implementation:

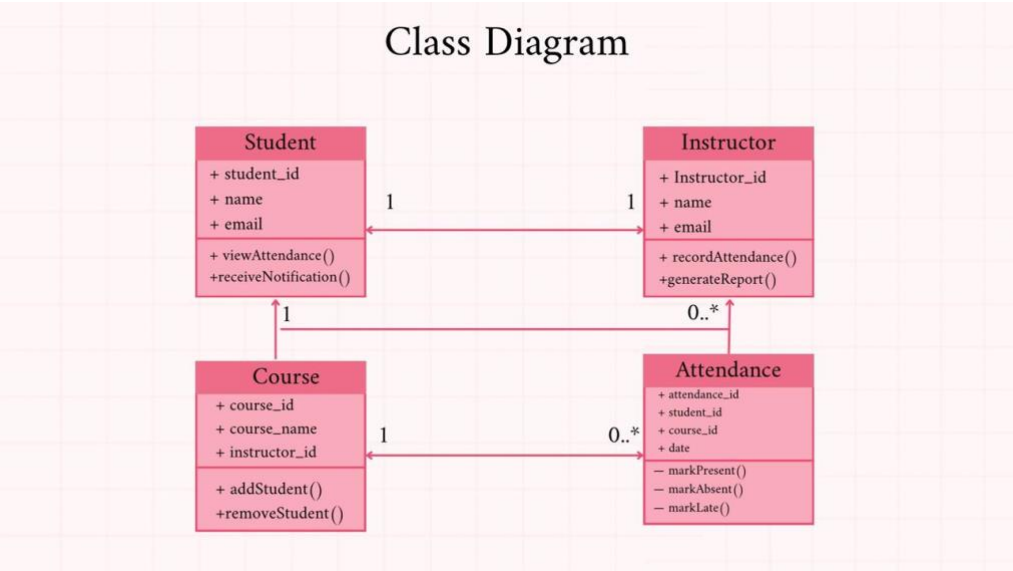
- **Use Case Diagram** was used to identify and describe the interactions between users (such as students, teachers, and administrators) and the system functions



- **Entity-Relationship (ER)** Diagram was used to model the database structure, define entities, and show the relationships between them.



- **Class Diagram** was used to represent the system structure in terms of classes, attributes, and methods in an object-oriented approach.



These diagrams helped in analyzing the system requirements clearly and improved the understanding of how the system should work before starting the coding phase.

7. Implementation

The implementation phase represents the actual development of the Student Attendance Management System using the Java programming language. The system was developed as a modular console-based application to ensure clarity, maintainability, and ease of understanding.

The system consists of two main classes:

1. Student Class

This class is responsible for representing the student entity in the system. It contains the student's information such as ID, name, and attendance count. It also includes methods to update and display attendance records.

2. Main Class

This class acts as the driver of the system. It provides a menu-driven interface that allows the user to perform different operations such as adding students, marking attendance, and viewing student records.

8. Testing

The system was tested using several testing strategies to ensure its correctness, reliability, and overall performance.

- Unit Testing: Each function in the system was tested individually to verify that it performs its intended task correctly. For example, functions such as adding a student, recording attendance, and displaying attendance were tested separately.
- Integration Testing: After testing individual components, the system was tested as a whole to ensure that all modules work together properly without any conflicts or errors.
- Dynamic Testing: The program was executed multiple times using different inputs to verify that it behaves correctly during runtime and produces accurate results.

In addition, different test cases were applied to check how the system handles various scenarios, such as entering valid and invalid data. This helped in identifying and fixing potential issues.

Overall, the testing process confirmed that the system functions as expected, and all features operate correctly and efficiently.

```
===== MENU =====
1. Add Student
2. Mark Attendance
3. View Students
4. Exit
Choose: 2
Enter Student ID: 112345
Attendance marked!

===== MENU =====
1. Add Student
2. Mark Attendance
3. View Students
4. Exit
Choose: 3
ID: 1134325 | Name: jood | Attendance: 0
ID: 112345 | Name: rana | Attendance: 1

===== MENU =====
1. Add Student
2. Mark Attendance
3. View Students
4. Exit
Choose: 4
Goodbye!
```

9. Deployment

The system was successfully deployed by uploading all project files to GitHub, making it accessible for sharing and version control. A readme.txt file was also included to provide clear instructions on how to run and use the system.

The deployment process ensured that the project is well-organized and easy to understand for other users or developers. It also allows future updates and improvements to be managed efficiently. Additionally, using GitHub helps in maintaining backup copies of the project and tracking any changes made during development.

The system was developed using Java as the main programming language and MySQL as the database management system. It was deployed as a local desktop application, allowing users to run it directly on their computers.

10. Conclusion

This project provided a practical understanding of how software engineering concepts can be applied to solve real-world problems. Starting from problem analysis to system

design, implementation, and testing, all phases of the Software Development Life Cycle (SDLC) were successfully followed.

Throughout the project, we learned the importance of proper planning, teamwork, and time management. We also gained experience in designing systems, writing code, and testing software to ensure its correctness and reliability.

Overall, the project enhanced our technical skills and gave us valuable experience in developing a complete software system.

11. Future Work

Although the current system meets the basic requirements, there are several improvements that can be implemented in the future:

- Developing a graphical user interface (GUI) to make the system more user-friendly and interactive.
- Enhancing the system by using a more advanced and scalable database solution for better performance and data handling.
- Adding a notification system (such as email or SMS) to inform students about their attendance status in real time.
- Developing a mobile application version of the system to allow access anytime and anywhere.

In addition, future enhancements may include improving system security through advanced authentication methods, adding more detailed and customizable reporting features, and expanding the system to support larger educational institutions with a higher number of users.

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